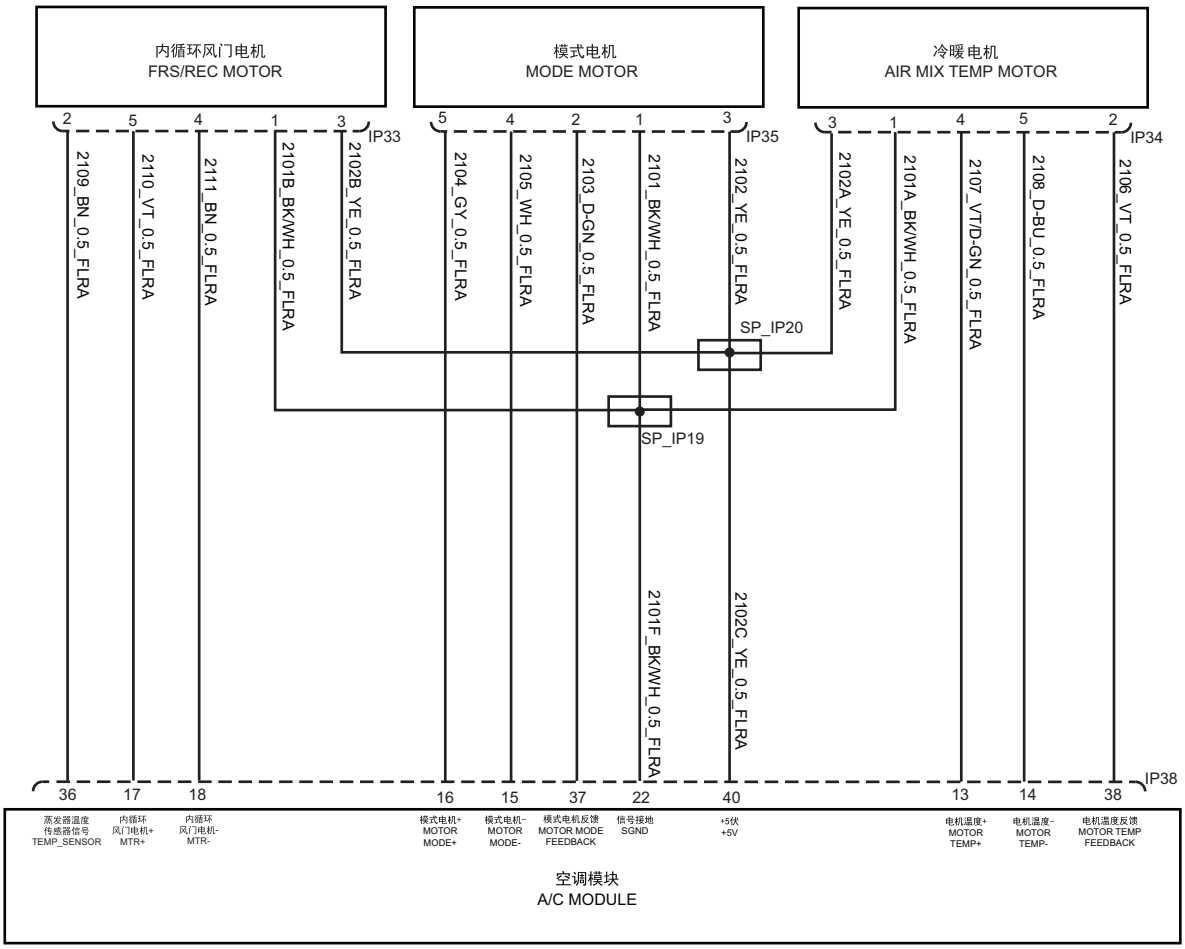
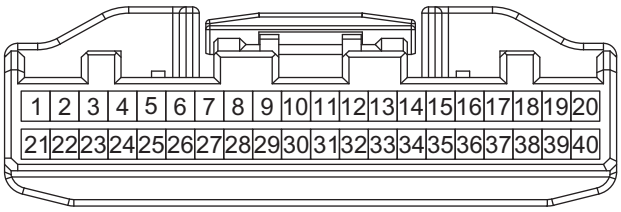
B1605-1C

Description of fault diagnosis codes
B1605-1C：Temp motor short or open
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • Air conditioning controller fault • Temp motor fault
Conditions for fault code operation
The ignition switch is in ON position.
Conditions for fault code setting
B1605-1C
To set the effect of a fault code condition
1 2
Description of circuit diagram
The temp motor provides power through the air conditioning controller, and the air conditioning controller module monitors and controls the mode motor by connecting wire harnesses.

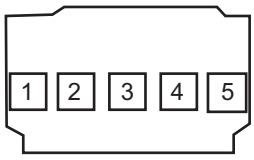
Circuit diagram



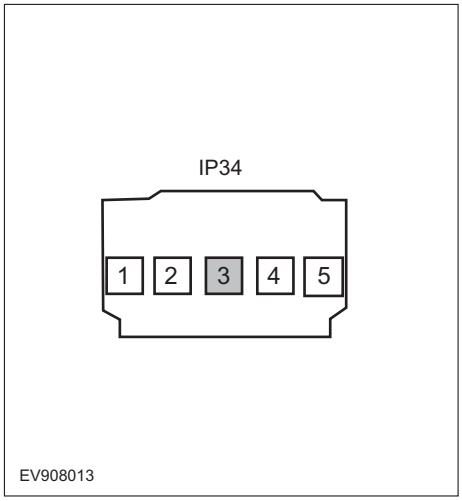
IP38

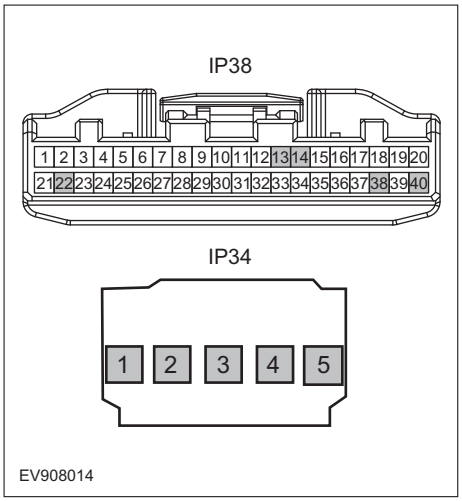


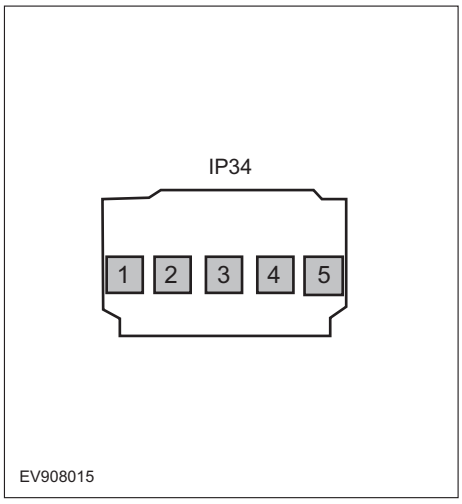
IP33/IP34/IP35

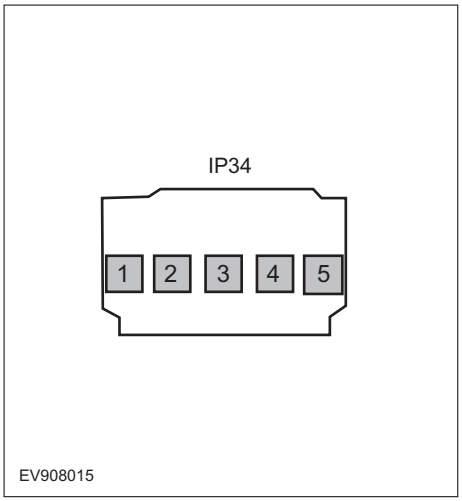


EV908009

Troubleshooting procedures	
Step 1	General inspection.
	1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of temp motor and air conditioning control module for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the air condition control system with diagnostic apparatus. 4 Is there any other fault code except for B1605-1C?
	YES → Refer to:DTC Summary List (HVAC).
NO ↓	
Step 3	Check the power supply circuit of temp motor circuit.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP34 of temp motor. 3 Turn the ignition switch to ON. 4 Measure the voltage between the terminal on harness plug of temp motor and ground. Measure circuit: voltage between terminal 3 on plug IP34 and ground. **Standard value:4.8-5.2V** 5 Check whether the result is normal or not?
	NO → Please repair or replace the faulted parts.
YES ↓	

Step 4	Check the circuit of temp motor (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP38 of air conditioning controller. 3 Measure the resistance between harness plugs of temp motor and of air conditioning controller. Measuring circuit: resistance between terminal 1 on plug IP34 and terminal 22 on plug IP38. Measuring circuit: resistance between terminal 2 on plug IP34 and terminal 38 on plug IP38. Measuring circuit: resistance between terminal 3 on plug IP34 and terminal 40 on plug IP38. Measuring circuit: resistance between terminal 4 on plug IP34 and terminal 13 on plug IP38. Measuring circuit: resistance between terminal 5 on plug IP34 and terminal 14 on plug IP38. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Please repair or replace the faulted parts.
YES ↓	

Step 5	Check the circuit of temp motor (short circuit).
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness plug of temp motor and ground. Measuring circuit: resistance between terminal 1 on plug IP34 and ground. Measuring circuit: resistance between terminal 2 on plug IP34 and ground. Measuring circuit: resistance between terminal 3 on plug IP34 and ground. Measuring circuit: resistance between terminal 4 on plug IP34 and ground. Measuring circuit: resistance between terminal 5 on plug IP34 and ground. **Standard value: ∞** 3 Measure the voltage between the harness plug of temp motor and ground. Measure circuit: voltage between terminal 1 on plug IP34 and ground. Measuring circuit: resistance between terminal 2 on plug IP34 and ground. Measuring circuit: resistance between terminal 3 on plug IP34 and ground. Measuring circuit: resistance between terminal 4 on plug IP34 and ground. Measuring circuit: resistance between terminal 5 on plug IP34 and ground. **Standard value: ≈ 0V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the temp motor (parts).
	1 Measure the resistance of temp motor terminal directly. Measuring circuit: measure the resistance between terminal 1 and 3 on plug IP34. **Standard value:** Measuring circuit: measure the resistance between terminal 4 and 5 on plug IP34. **Standard value:** 2 Measure the resistance between the temp motor terminal and the housing. Measuring circuit: Measure the resistance between the terminal 1 on plug IP34 and the housing. Measuring circuit: Measure the resistance between the terminal 2 on plug IP34 and the housing. Measuring circuit: Measure the resistance between the terminal 3 on plug IP34 and the housing. Measuring circuit: Measure the resistance between the terminal 4 on plug IP34 and the housing. Measuring circuit: Measure the resistance between the terminal 5 on plug IP34 and the housing. **Standard value: ∞** 3 Check whether the result is normal or not?
	NO → Change the temp motor.
YES ↓	
Step 7	Check if air conditioning module is normal.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug IP38 of air conditioning controller. 3 Check and repair; • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the air conditioning control module plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the air conditioning control module.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	

Next Step ↓
Diagnosis end.